



Drugs, Indications, and Side Effects

Network Science Project
University of Padova - MSc Data Science
Elizaveta Kuzovkova

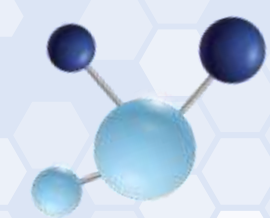




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01

Introduction



Background, objectives, research questions



Background

What is a drug network?

Medications can share biological and clinical characteristics. Connections between drugs can be inferred from common therapeutic uses, similar mechanisms of action, reported adverse effects, molecular or gene structures, and other properties.

The complex set of such relationships can be represented in the form of a network.

Why study them?

Drug networks provide a systematic view on pharmacology that is difficult to obtain from individual drug analysis. The network science approach helps to uncover hidden relationships, which are used for solving novel problems: drug repurposing, drug interaction prediction, side effects detection, etc.





Objectives

- Focus on Nervous System pharmacological group of medications
 - Construct a drug similarity network based on shared side effects profiles
 - Construct a drug similarity network based on shared therapeutic indications
 - Compare network structures and communities
 - Evaluate how side effect data can reveal pharmacological structure
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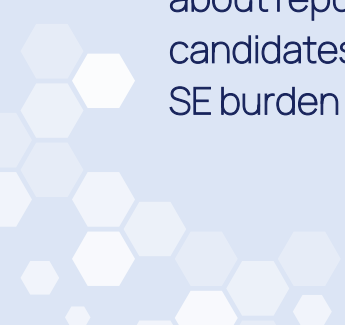
Research questions



01

Centrality

Which drugs hold the central positions in each network, what this reveals about repurposing candidates and greatest SE burden items?



02

Clusters

How drugs with similar indications and SE profiles group in the network, and how do these communities relate to ATC classes?

03

Assortativity

Are nervous system drugs assortative by ATC groups and subgroups?

Key concepts



Anatomical Therapeutic Chemical (ATC) Classification

Medical substances system implemented by World Health Organization. The 1st level characterizes the main therapeutic use.



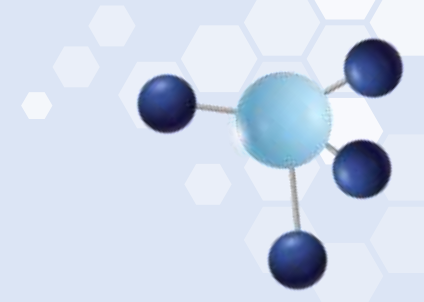
Side Effect (Adverse Effect)

An unintended effect of medication reported during clinical use.



Drug Indication

The approved therapeutic use of a drug for a specific condition.



02

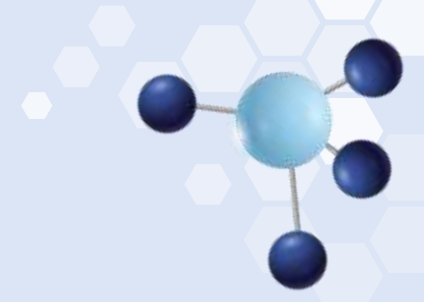
Literature review





Literature review

- Pavlopoulos, G. A., et al. (2018). *Bipartite graphs in systems biology and medicine: a survey of methods and applications*. GigaScience, 7(4), giy014.
Introduces the usage of bipartite graphs for biological data, including biomedical examples of diseases, drugs, adverse drug effects. Also covers methods for bipartite and unipartite projection network topology analysis.
 - Cavallo, P., et al. (2013). *Network analysis of drug prescriptions*. Pharmacoepidemiology and drug safety, 22(2), 130-137.
Analyzes co-prescription drug network at various ATC levels, uncovering link patterns corresponding to patients' age and gender. For example, the younger age shows stronger links to antibiotics group (may be due to infective diseases prevalence), while for the older one the central role is taken by cardiovascular system drugs.
 - Keng, Y. Y., Kwa, K. H., & Ratnavelu, K. (2021). *Centrality analysis in a drug network and its application to drug repositioning*. Applied Mathematics and Computation, 395, 125870.
Constructs a network of drugs by side effects similarities, calculate medications centralities and evaluate a model for prediction of drug repositioning based on the centralities measures.
 - Askar, M., et al. (2021). *An introduction to network analysis for studies of medication use*. Research in Social and Administrative Pharmacy, 17(12), 2054-2061.
Examines co-medication, severe drug-drug interactions, combined networks through topological, assortativity, and modularity analysis.
- 



03

Data description



Datasets



repoDB

Catalogue of medications on various stages of approval and their indications, extracted from DrugCentral and ClinicalTrials.gov



SIDER

Marketed medicines and their reported adverse drug reactions, extracted from package inserts and public regulatory documents



Preprocessing and exploration

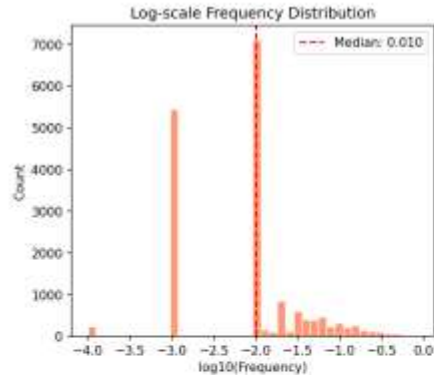
Cleaning & matching

- Drug name normalization
- Exclude placebos, all drugs that are not approved for use
- Filter to ATC code N for nervous system
- Side effects: take max frequency of reported trials, placeholders for frequency -> set to min detectable frequency (0.001)

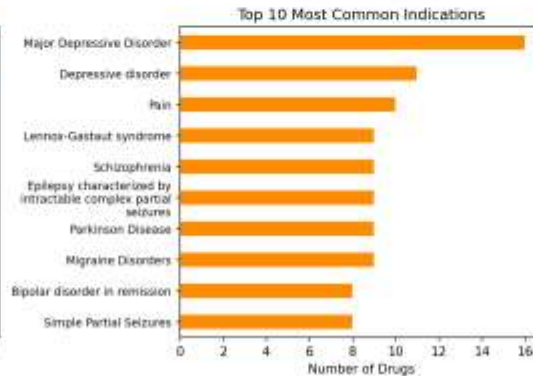
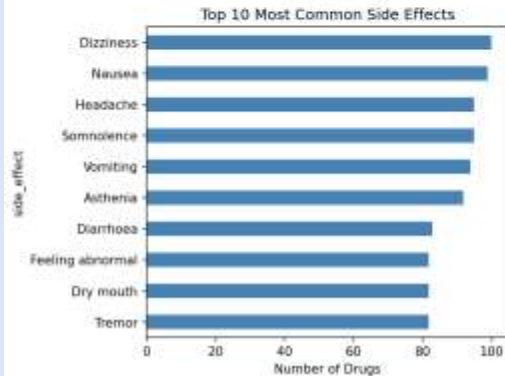
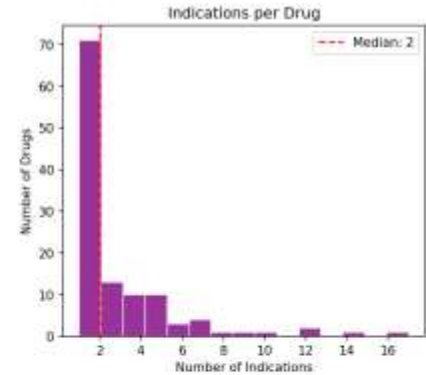
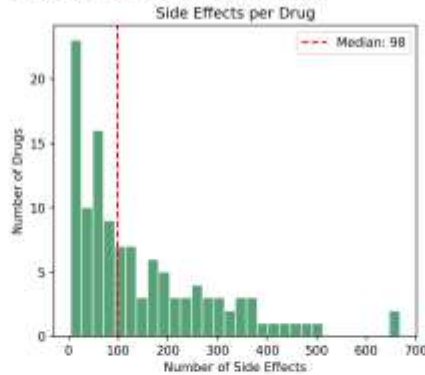
Exploration

- 118 drugs total
- 17 304 unique drug-SE pairs
- 352 unique drug-indication pairs
- 7 ATC2 classes (therapeutic subgroup)
- 17 ATC3 classes (pharmacological subgroup)
- 46 ATC4 classes (chemical subgroup)

Data exploration



Drug Network Data Exploration



- Drug-Drug Indication is assortative (0.2370)
- Other networks are disassortative or closer to neutral



04

Networks statistics



Networks construction



Drug-Indication

- Bipartite
- Nodes: 250
- Edges: 352
- Density: 0.0113

Drug-SE

- Bipartite weighted by SE frequency
- Nodes: 1901
- Edges: 17304
- Density: 0.0096

Drug-Drug by indication

- Projection, weighted by number of shared indications
- Nodes: 118
- Edges: 552
- Density: 0.0800

Drug-Drug by SE count

- Projection, weighted by number of shared SE
- Nodes: 118
- Edges: 6731
- Density: 0.9751

Drug-Drug by SE Jaccard

- Projection, accounts for SE profiles
- After sparsification:
- Nodes: 118
 - Edges: 3363
 - Density: 0.4872

Drug-Drug by SE Cosine

- Projection, accounts for frequencies matches
- After sparsification:
- Nodes: 118
 - Edges: 875
 - Density: 0.1268



Networks properties

Drug-Indication

- Connected components: 8
- Largest: 232 (92.8%)
- Clust. coef: 0.36
- Median degree: 2
- Max degree: 17

Drug-SE

- Connected components: 1
- Largest: 1901 (100.0%)
- Clust. coef: 0.14
- Median degree: 3
- Max degree: 668

Drug-Drug by indication

- Connected components: 8
- Largest: 108 (91.5%)
- Clust. coef: 0.71
- Median degree: 9
- Max degree: 29

Drug-Drug by SE count

- Connected components: 1
- Largest: 118 (100.0%)
- Clust. coef: 0.98
- Median degree: 117
- Max degree: 117

Drug-Drug by SE Jaccard

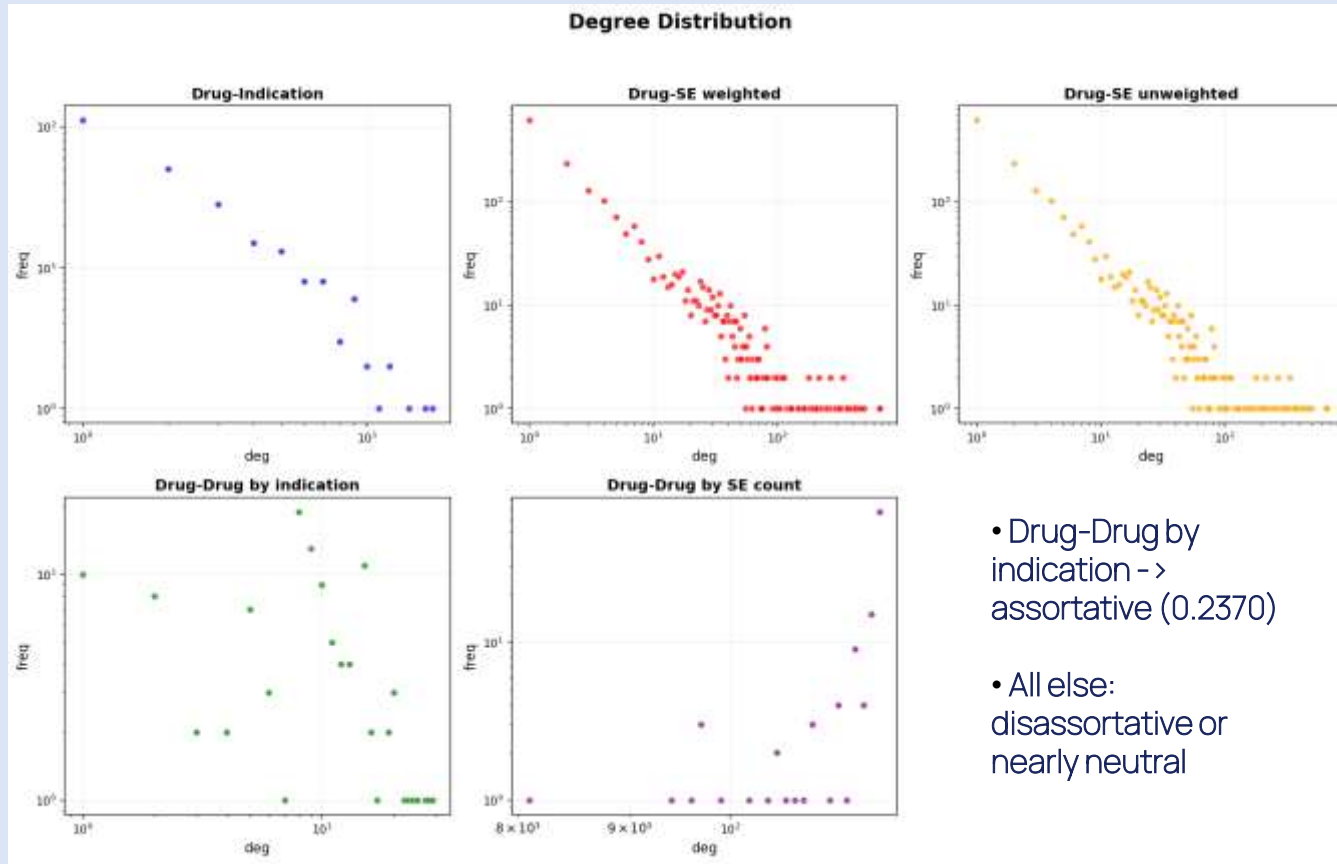
- After sparsification:
- Connected components: 12
 - Largest: 103 (87.3%)
 - Clust. coef: 0.73
 - Median degree: 72
 - Max degree: 92

Drug-Drug by SE Cosine

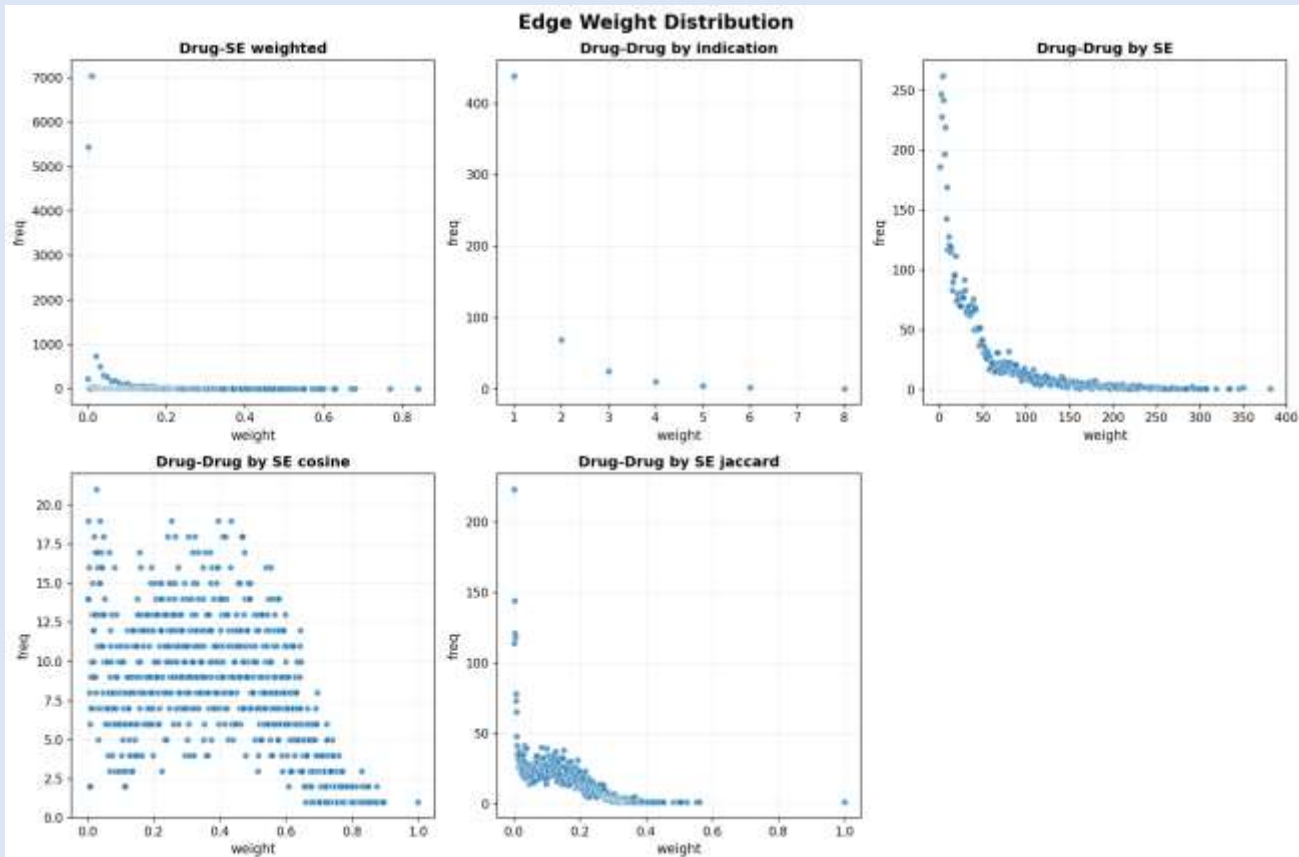
- After sparsification:
- Connected components: 1
 - Largest: 118 (100.0%)
 - Clust. coef: 0.39
 - Median degree: 14
 - Max degree: 30



Degree distribution



Edge weight distribution





05

Centrality analysis



Top indications



Degree

- Major Depressive Disorder: 0.1356
- Depressive disorder: 0.0932
- Pain: 0.0847
- Parkinson Disease: 0.0763
- Lennox-Gastaut syndrome: 0.0763

-> most targeted conditions

Betweenness

- Postherpetic neuralgia: 0.1737
- Muscle Spasticity: 0.1701
- Major Depressive Disorder: 0.1697
- Severe pain: 0.1322
- Restless Legs Syndrome: 0.1248

-> conditions that bridge separate drug groups

PageRank

- Major Depressive Disorder: 0.0205
- Parkinson Disease: 0.0143
- Migraine Disorders: 0.0143
- Depressive disorder: 0.0132
- Pain: 0.0122

-> conditions treated by the most central drugs

Top side effects



Cumulative frequency

- Somnolence: 18.7232
- Nausea: 18.4739
- Headache: 15.2926
- Dizziness: 14.8756
- Asthenia: 10.7358

-> heaviest overall burden by reported frequency

Degree

- Dizziness: 0.8475
- Nausea: 0.8390
- Headache: 0.8051
- Somnolence: 0.8051
- Vomiting: 0.7966

-> most universal adverse reactions

PageRank

- Nausea: 0.0167
- Somnolence: 0.0164
- Headache: 0.0136
- Dizziness: 0.0134
- Asthenia: 0.0095

-> SE carried by many central drugs



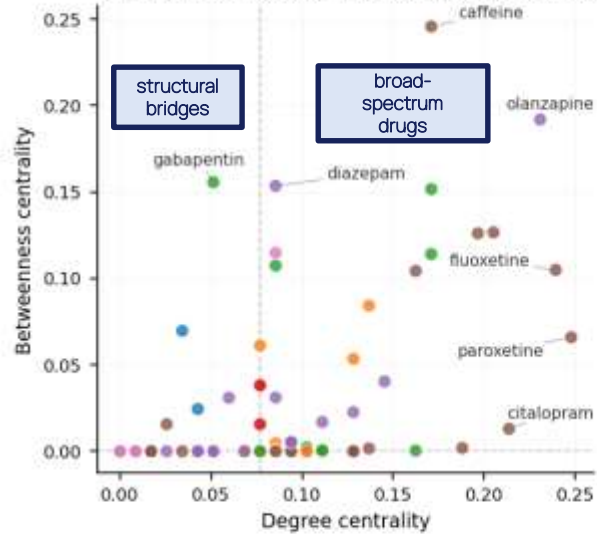
Measures & interpretations

Network	Measure	Interpretation
Drug-Drug by Indication	Degree	Has a variety of therapeutic uses shared by other medications
	Strength	Sits in a dense therapeutic neighborhood
	Betweenness	Bridges distinct indication clusters
	PageRank	Influential in the network, potential popular baseline medication
Drug-Drug by SE cosine	Degree	Has similar SE profiles to many others, non-specific
	Strength	Broad and overlapping adverse effect pattern
	Betweenness	Connects different SE clusters, spans multiple pharmacological subclasses
	PageRank	Has highly replicated SE pattern characteristic of major drug classes

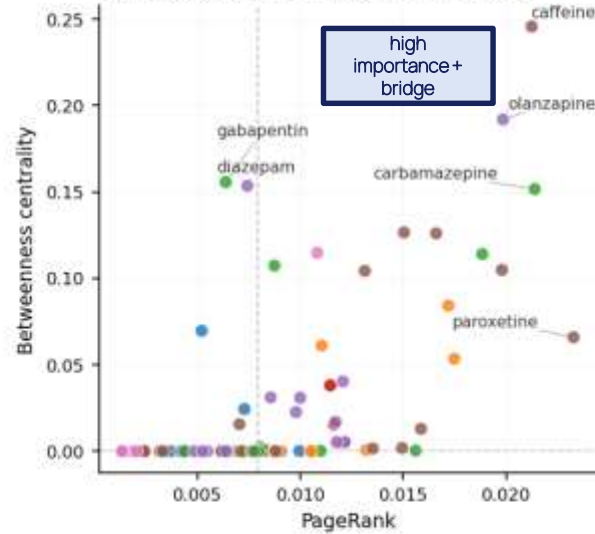
Indications network



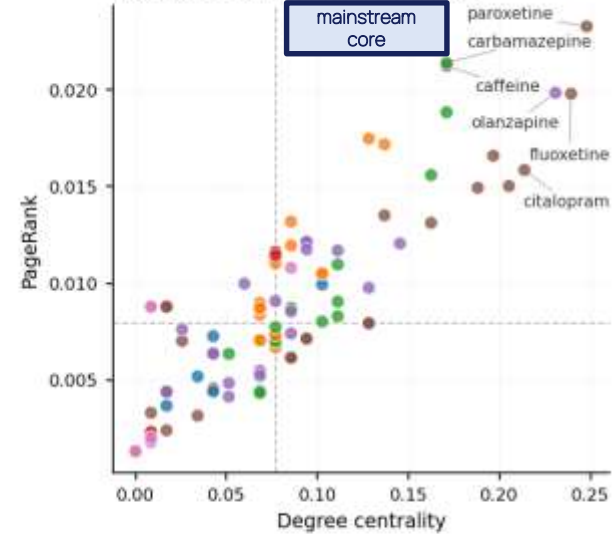
Drug-Drug Indication
Degree centrality vs Betweenness centrality



Drug-Drug Indication
PageRank vs Betweenness centrality

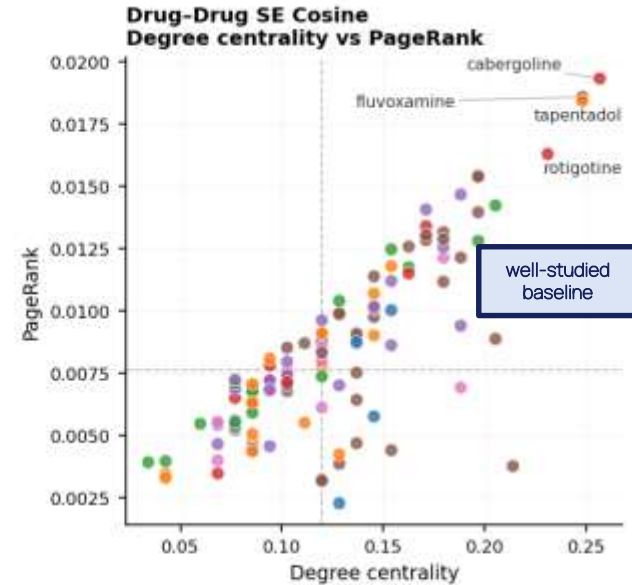
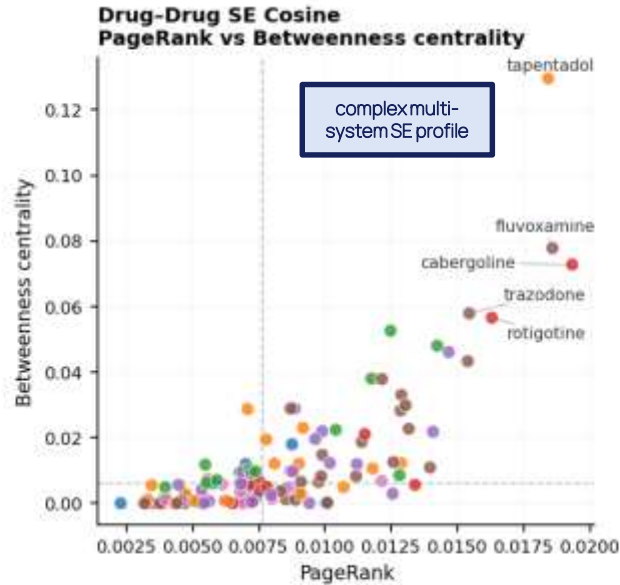
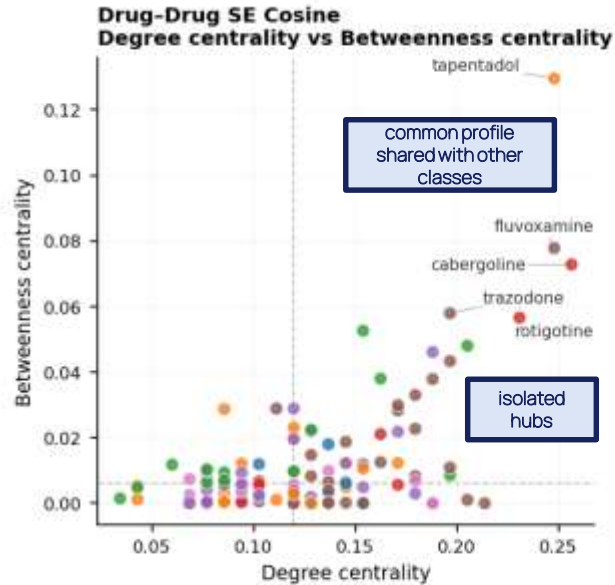


Drug-Drug Indication
Degree centrality vs PageRank



ATC2
N01 N02 N03 N04 N05 N06 N07

SE network



Example of finding replacements

Target

Medication with the highest total SE burden: risperidone

Top replacements*

drug	indication_overlap	se_burden	cosine_se_similarity	score
aripiprazole	5	4.97	0.580	0.3120
ziprasidone	4	4.02	0.474	0.2668
carbamazepine	3	4.49	0.354	0.1787
asenapine	3	2.05	0.469	0.1711
olanzapine	4	10.04	0.465	0.1653
quetiapine	3	7.79	0.570	0.0926

01

Indication match

$$\left(\frac{\text{ind_sim}}{\text{max_overlap}} \right)^2$$

Maximized

02

SE burden

$$\left(1 - \frac{\text{burden}}{\text{anchor_burden}} \right)$$

Minimized

03

SE similarity

$$(1 - \text{cosine_sim})$$

Minimized

Scoring

* Some studies show that Aripiprazole can indeed be more favorable replacement in the sense of tolerability (Ishitobi, M. et al., 2012), effectiveness (Rezayat, A. et al., 2014), and short-term metabolic profile (Rahimi Darehbagh, R., 2025).



06

SE network sparsification

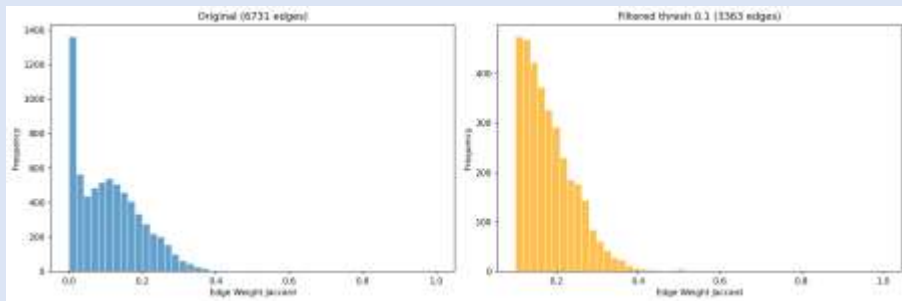




Approaches overview

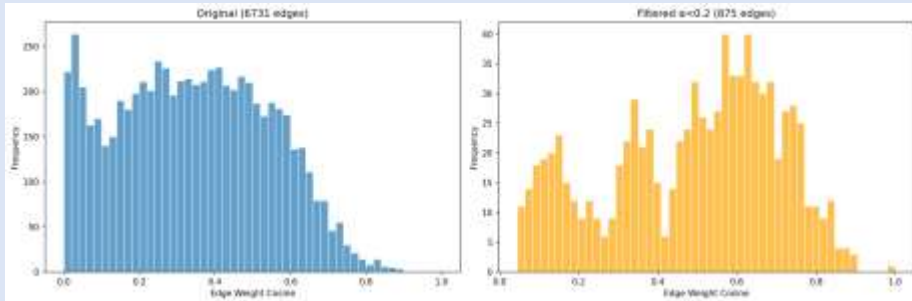
Drug-Drug by SE Jaccard

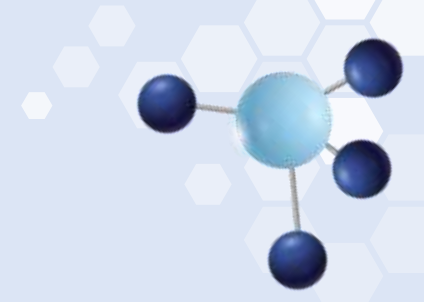
- Arbitrary threshold on edge weight (drop if weight < 0.1)
- Edges: 6731 -> 3363
- Density: 0.9751 -> 0.4872



Drug-Drug by SE Cosine

- Disparity filter algorithm of weighted networks (from Serrano, et al., 2009)
- Statistical significance test of edges against null model (random assignment from a uniform distribution)
- Edges: 6731 -> 875
- Density: 0.9751 -> 0.1268





07

Community detection



Algorithms tested

Indications network

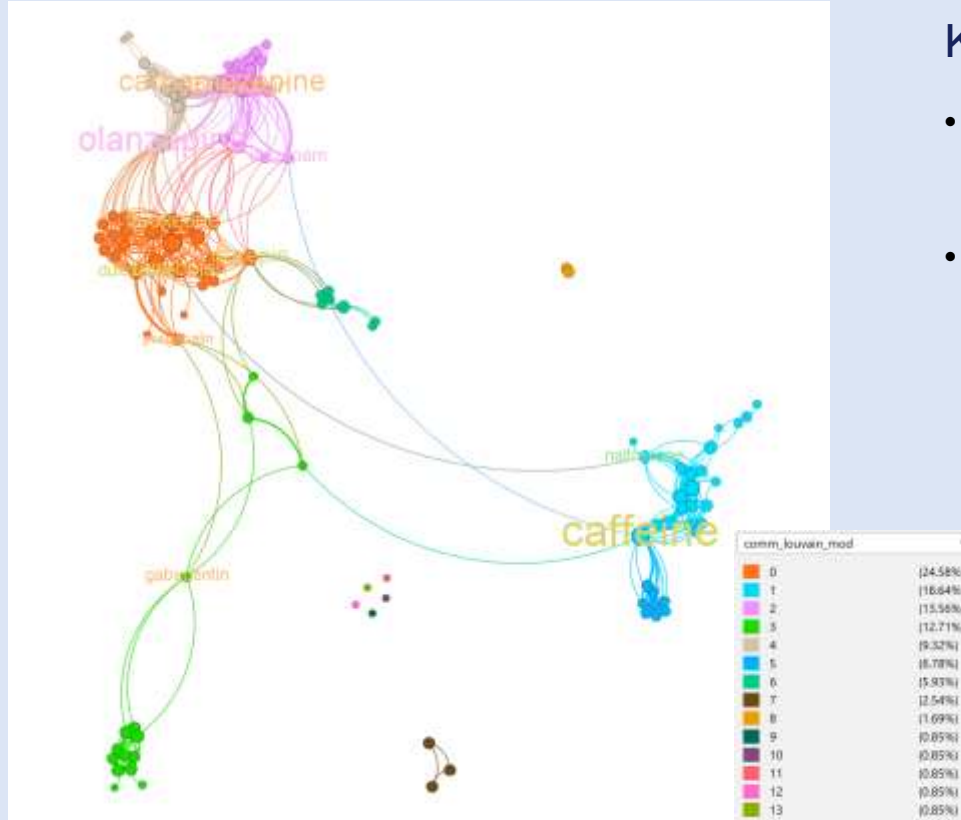
- Louvain (Mod): 14 communities, modularity=0.6743, Ncut=0.0979
- Louvain (RB): 14 communities, modularity=0.6743, Ncut=0.0979
- Louvain (CPM): 22 communities, modularity=0.6393, Ncut=0.4947
- Infomap: 16 communities, modularity=0.6716, Ncut=0.1515
- Edge Betweenness: 12 communities, modularity=0.6215, Ncut=0.0608

SE network

- Louvain (Mod): 5 communities, modularity=0.4642, Ncut=0.3427
- Louvain (RB): 5 communities, modularity=0.4642, Ncut=0.3427
- Louvain (CPM): 19 communities, modularity=0.4275, Ncut=0.7505
- Infomap: 7 communities, modularity=0.4591, Ncut=0.4199

Main approach -> Louvain with ModularityVertexPartition

Indications network



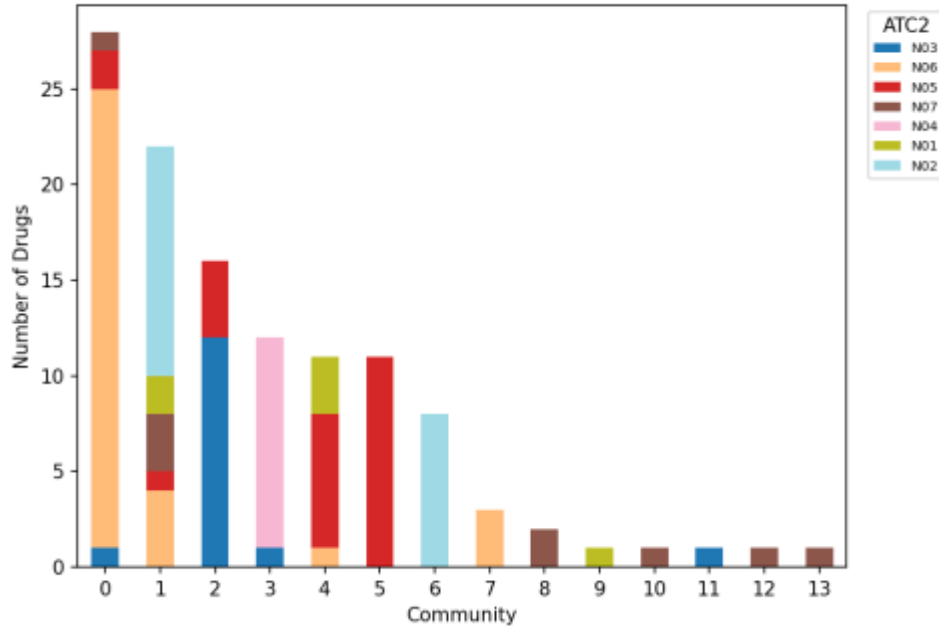
Key observations

- Community sizes: [29, 22, 16, 15, 11, 8, 7, 3, 2, 1, 1, 1, 1, 1]
- Antiepileptics (N03) and Psychoanaleptics (N06) groups are the main bridges between communities (highlighted by betweenness orange and yellow node labels respectively)

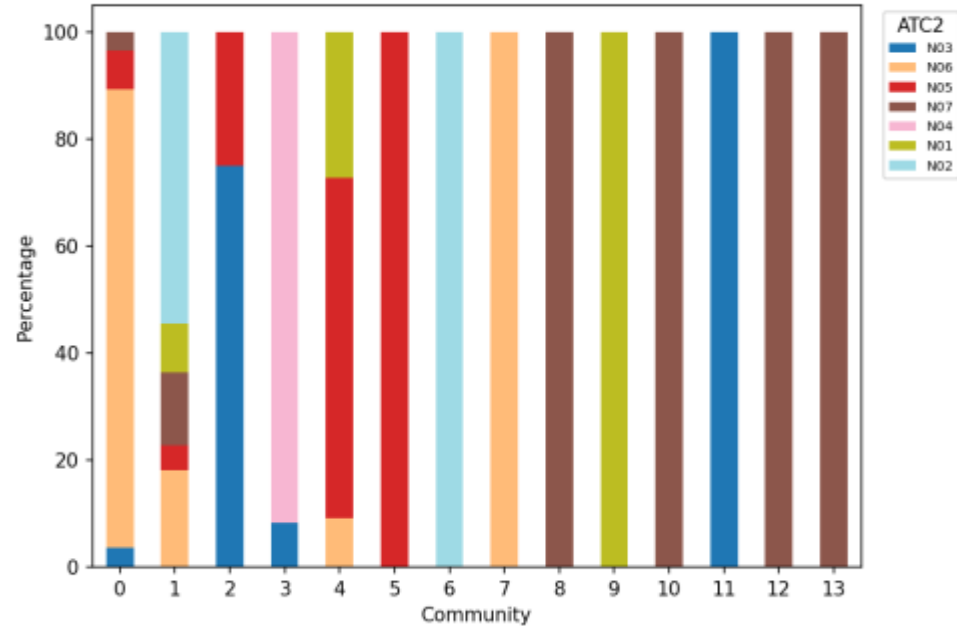
Indications network



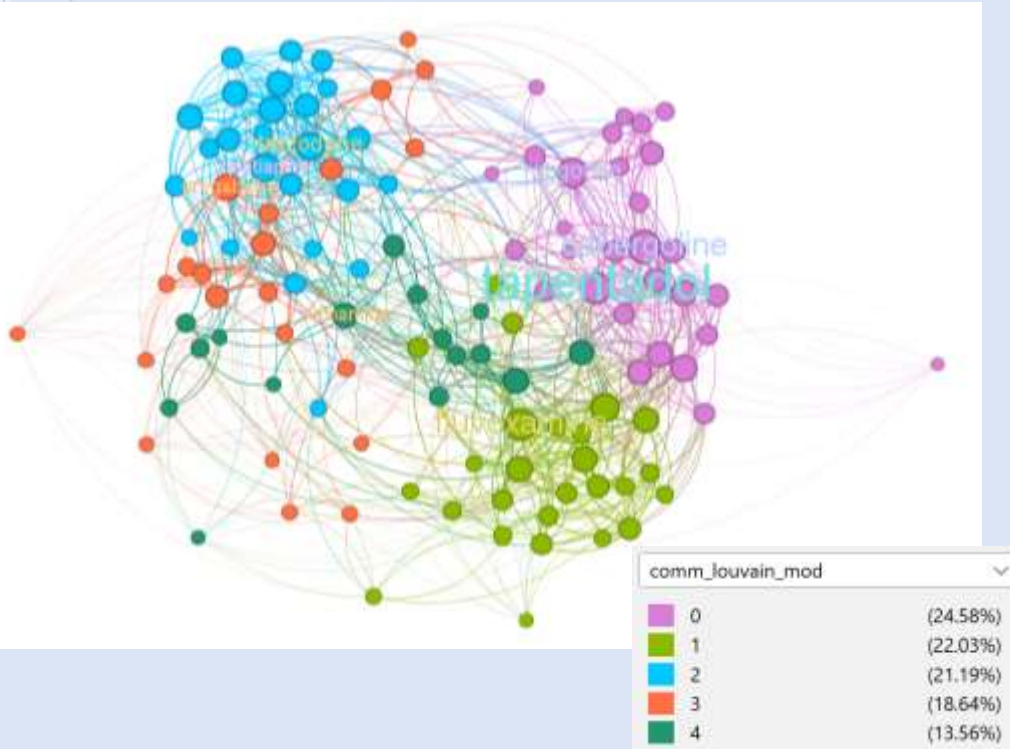
Drug-Drug by Indication: ATC2 (counts)



Drug-Drug by Indication: ATC2 (%)



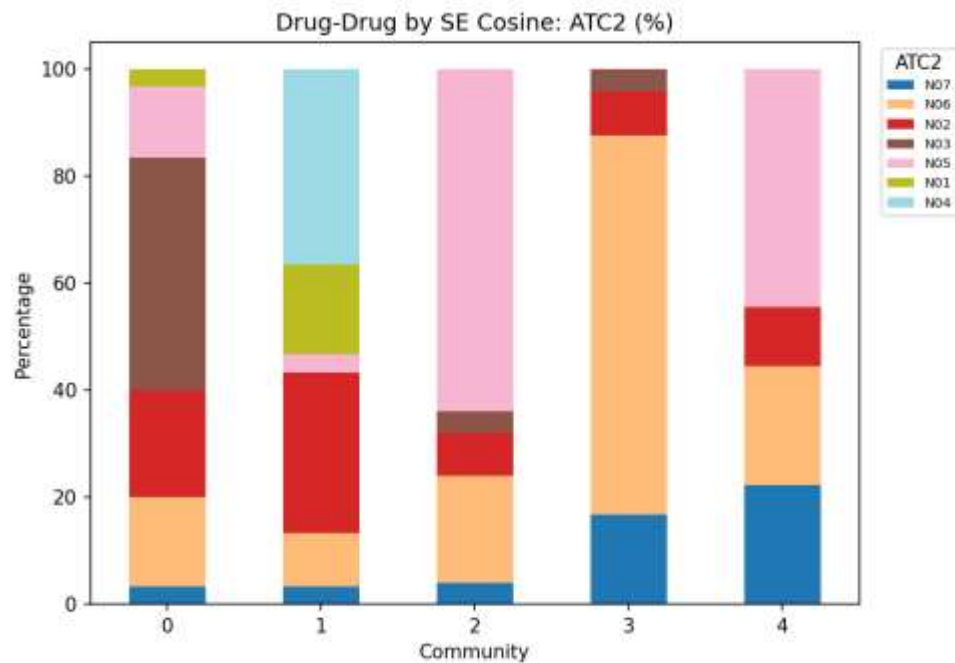
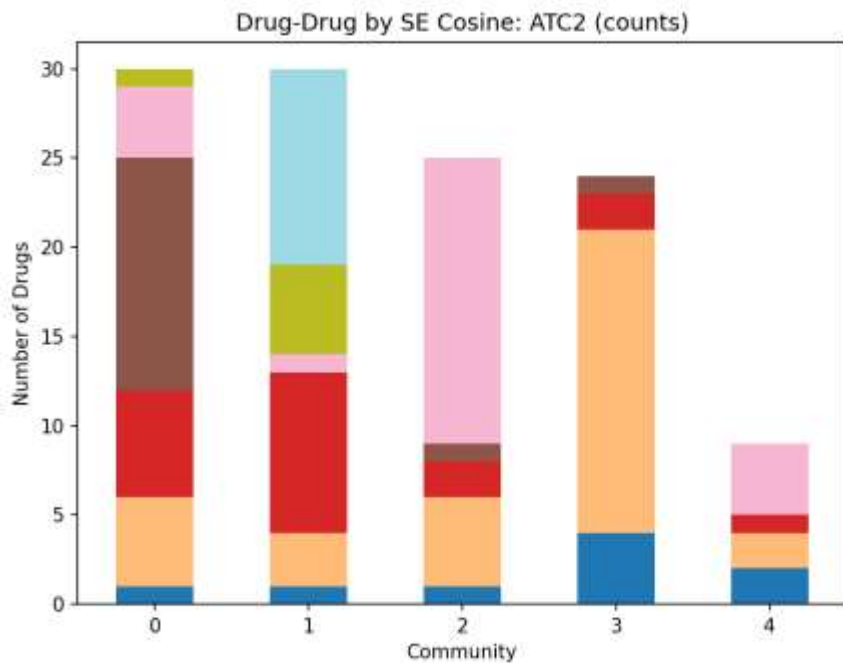
SE network



Key observations

- Community sizes: [30, 30, 25, 24, 9]
- No isolated communities/nodes, still a dense network
- Tapentadol (Analgesics N02, opioid) stands out in PageRank (node size) and betweenness (label highlight) -> broad and significant toxicity profile
- Some Antiepileptics (N03, orange label), Anti-parkinson (N04, blue), and Psychoanaleptics (N06, yellow) drugs connect various SE profiles clusters

SE network





08

Assortativity analysis

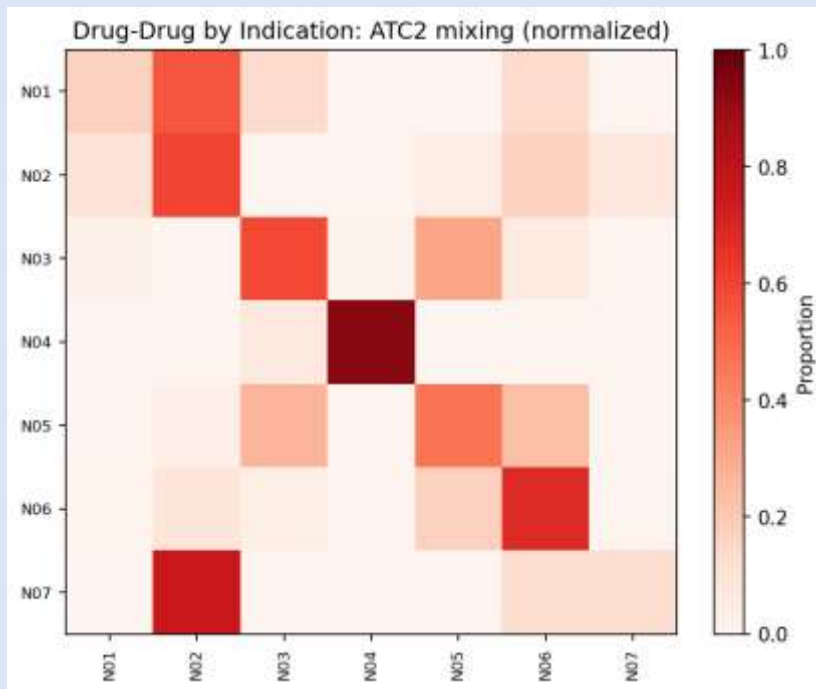


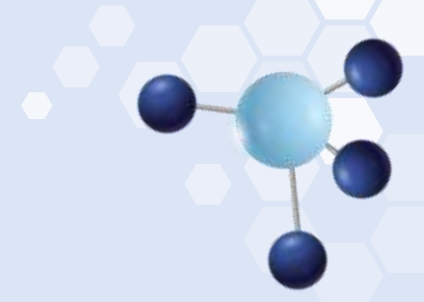


Indications network

Key observations

- On more granular ATC level (from ATC2) assortativity rises: +0.6419
- 58.7% of edges are within-class, most drug classes strongly connect to themselves
- Analgesics (N02) also tends to connect to Anesthetics (N01) and Other nervous system drugs (N07)





09

Conclusion



Answers to research questions

01

Centrality

Which drugs are central, and what does it reveal about repurposing and SE burden?

Indications: hubs are broad-use CNS drugs (as caffeine) with high therapeutic flexibility;

SE: a different set leads (tapentadol, fluvoxamine, cabergoline): these medications have the most typical adverse profiles.

02

Clusters

How do similar drugs group, and do communities map to ATC classes?

Indications: clean community structure is present with some bridges, most clusters are dominated by one ATC2 group;

SE: structure is fuzzier, still dense without any isolated nodes, smaller number of communities lead to bigger variety in ATC2 groups.

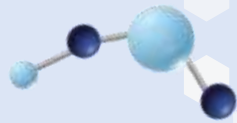
03

Assortativity

Are Nervous system drugs assortative by ATC group?

Indications: yes, assortative by ATC2 and more granular ATC3 level, therapeutic class drives indication similarity;

SE: close to neutral, adverse effects profile similarity crosses class boundaries.



Discussion

01

Projection choice

Raw SE count projection is near-complete graph, only after applying disparity filter network becomes interpretable.

02

Bridges matter

The most interesting nodes are the betweenness outliers, not dominating hubs, as they reveal cross-class connections.

03

Opposite leaders

The central drugs between indications and SE networks barely overlap.

04

Role of metrics

Degree and PageRank are nearly collinear in the drug-drug projections and the bipartite networks require specific normalization for valid rankings.

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